Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/03/2006
PI Name:	Cheng, Robert K. Ph.D.		
Project Title:	Field Effects of Gravity on Lean Premixed Turbulent Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous flames - premixed		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720	**Congressional District:**	7
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	97-HEDS-01
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	3		
COI Name (Institution):			
Grant/Contract No.:	C32121A		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		

This task is focused on experimental reserach to quantiry the effects of buoyancy on the mean flowfield and its impact on local and overall flame/turbulence interaction processes. The approach is to obtain detailed measurement and analysis of the velocity and scalar fields of premixed turbulent flames in microgravity and in normal, partial and enhanced gravity.

The objective of this research is to proof that buoyancy is a system parameter of laboratory and practical premixed turbulent flames. Buoyancy provides an upstream feedback mechanis that dictates the overall flame flowfield as well as locak flame turbulence interaction processes. Consequently the flame structures, stabilization limits, turbulent reaction rates are directly or indirectly coupled with buoyancy.

Task Description:	The subject of our reserach, lean premixed turbulent flames, is an emerging environmental energy technology because these flames can be exploited to reduce the emission of oxides of nitrogen. Therefore, tesults of this research will benefit both combustion science and technology. Combustion engineer will have scaling information to eliminate buoyancy influences on practical systems. Experimentalists will gain new knowledge to identify and interpret buoyancy contributions to laboratory flames. Theoretical research will benefit by having a set of reference data to improve the fidelity of turbulent combustion models and to develop methods to address elliptical problems. Specifically, this research effort will generate the insight to determine significant processes through which gravity affects flame properties such as flame speed and flame structures. The results can be applied to investigate the gravity-influenced limit for premixed turbulent flames. The backward coupling effects of buoyancy (i.e. downstream effects felt upstream) is of particular importance to combustion theory because it relates to the elliptical problem that requires full time dependent treatment of upstream, wall and downstream boundary conditions. Knowledge on the back-coupling buoyancy affected processes is valuable for guiding the development of turbulent combustion models to include these effects. This study will contribute to resolving the inconsistencies that exist between experiments and theories.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Laboratory study of CLEAN flame is an important aspect of our research that provides background information necessary to interpret data from the flight experiments. We have completed Particle Image Velocimetry (PIV) measurements of three laminar and three turbulent CH4/air flames in +1g and -1g under the same initial conditions as those in our flight matrix. The experiments were performed at equivalence ratios, = 0.7, 0.8 and 0.9 with the mean flow velocity U at 3 m/s. For the turbulent flames, a perforated plate was used and it produced a slightly anisotropic turbulence with the axial rms velocity u' = 0.44 m/s and radial rms velocity v' = 0.3 m/s. For each flame, the PIV data have been analyzed to obtained the axial and transverse man and rms velocities. Comparing the profiles of +1g and -1g flames provided additional insights into the flowfield characteristics of the CLEAN flame burner and show that the effects of buoyancy occur not only in the flame flowfield but also in the wake region of the flame stabilizer. This is a clear demonstration of the upstream coupling due to field effects of gravity. However, the limited studies available in the open literature on the characteristics of the wake regions downstream of flame stabilizer are not sufficient to help gain better understanding of the processes that cause such significant differences. We also co-ordinate our work with those of Ian Shepherd's NASA project on "Numerical Investigation of Buoyancy and Pressure Affects in Premixed Flames." The objective of his work is to develop a numerical methodology to address and to be able to simulate the large difference between the length scales that characterize the combustion and those that characterize the buoyant interactions. For their first problem, Shepherd and Day simulated in 2D the unstable plume of +1g and -1g conical flames reported in one of our earlier studies. During FY04, they adapted the AMR method to simulate again in 2D a buoyancy stabilized laminar flames reported in another of our publications. Their current plan is to extend the scope of their investigation to simulate additional conical flames to evaluate and refine their simulation methods. Our goal is to facilitate the merging of experiments and simulation through a synergistic effort so that the computational method can be advanced the capability of the AMR methods to simulate in 3D time dependent processes of buoyant and microgravity CLEAN flames. We have also collaborated on the microgravity combustion project of Prof. Toshihisa Ueda of Keio University. Professor Ueda's research is to study the relative effects of buoyancy and solid body rotation on the instability of Bunsen type conical flames. The foundation of his project was our previous work on measuring and correlating pulsation frequency of buoyant conical flames. The objective is to extend the investigation to include the study of centrifugal forces on buoyancy effects. Our collaboration has produced two publications.
Bibliography Type:	Description: (Last Updated: 07/09/2019)
Articles in Peer-reviewed Journals	Gotoda H, Ueda T, Cheng RK. "Dynamic motion of rotating Bunsen flame tip in microgravity." AIAA Journal. 2004 Jul;42(7):1485-89. https://doi.org/10.2514/1.2502 , Jul-2004